ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Physical Insights and Diagnostic Analysis of the KG Formula

I. Physical Insights into the KG Trace Formula

II. Spectral Dynamics and Quasiparticle Lifetime Analysis

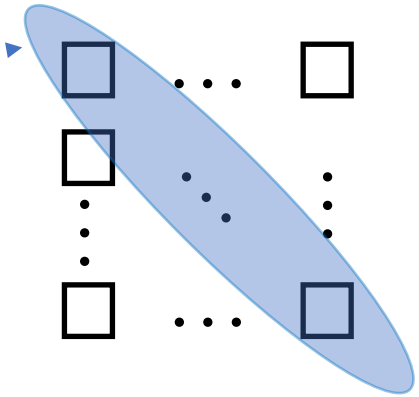
III. Numerical Diagnostics and Convergence in Computational Transport

I. Physical Insights into the KG Trace Formula

Kubo-Greenwood Formula $\sigma_{\mu\nu}(\omega) = \frac{\pi e^2 \hbar}{V} \int dE \left(-\frac{df}{dE}\right) Tr\{v_\mu [G_n^R(E) - G_n^A(E)] v_\nu [G_m^R(E) - G_m^A(E)]\}$

What the formula reveals about the system?

The Trace (Tr{…}) as an ensemble sum over all possible electronic states



Calculation of the probability of a particle transitioning between all available energy states at E_F, weighted by their connectivity, rather than a simple average

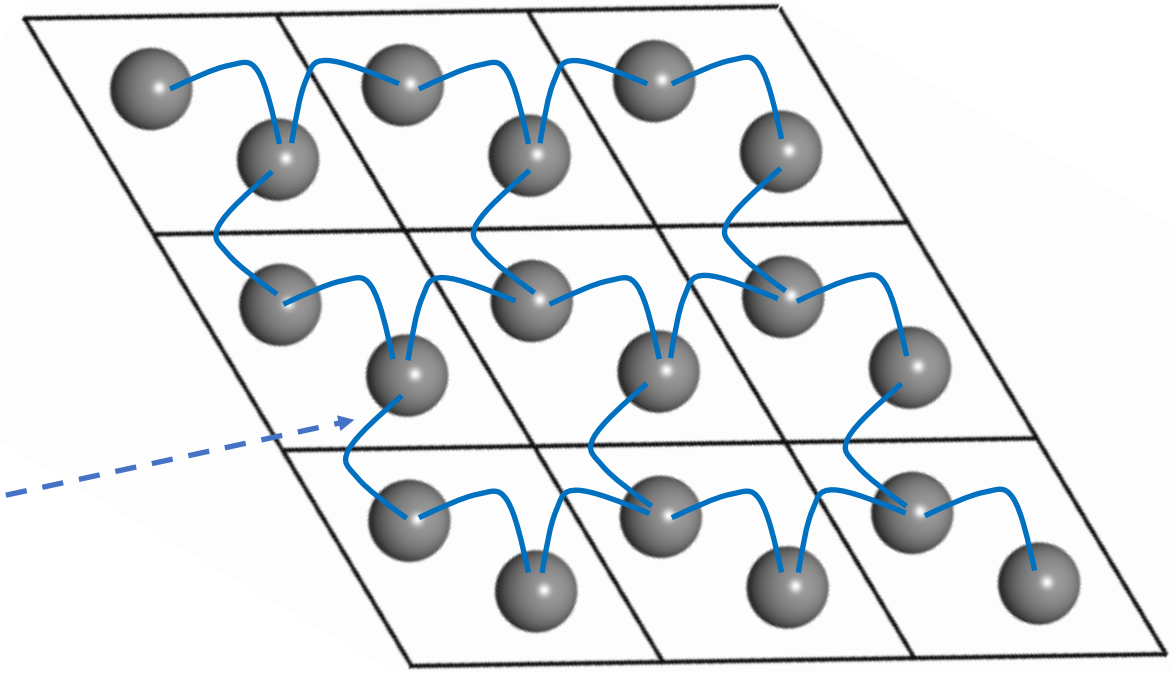
I. Physical Insights into the KG Trace Formula

In the Kubo-Greenwood (KG) framework, conductivity is re-conceptualized as a measure of the lattice connectivity. The lattice is viewed as a complex quantum network where atomic orbitals are nodes and hopping integrals (t_{ij}) act as the connecting 'edges'. The KG formula quantifies the ability of an electronic wave function to propagate coherently across this network at a specific Fermi energy (E_F)

wave function of electron



Jumping between each atoms



All electrons moving in the same way at same energy

charge current

I. Physical Insights into the KG Trace Formula

Details about the green function

Schrödinger Picture: wave function changes with time; while, operator does not $|\psi(t)\rangle = e^{-i(t-t_0)E_n/\hbar}|\psi(t_0)\rangle$

Time evolution operator: $U(t,t_0) = e^{-i(t-t_0)H/\hbar}$

The expectation value: $\langle r|U(t,t_0)|r'\rangle = \langle r|e^{-i(t-t')H/\hbar}|r'\rangle$

Suppose a charge current applied to the system at position r' **and time instant** t' → **The system response to this stimuli at position** r **and time instant** t

Retarded green function: $G^R(r,t,r',t') = -i\Theta(t-t')\langle r|e^{-i(t-t')H/\hbar}|r'\rangle$ $\quad \Theta(t-t') = \begin{cases} 0, \text{ when } t < t' \\ 1, \text{ when } t \geq t' \end{cases}$

Propagate the stimuli applied to the system at position r' **and time instant** t' **to position** r **and time instant** t

Retarded green function is also named as Propagator Or Response function

II. Spectral Dynamics and Quasiparticle Lifetime Analysis

$\tau = t - t'$ Retarded green function: $G^R(\tau) = -i\Theta(\tau)\langle r|e^{-i\tau H/\hbar}|r'\rangle$

Fourier transform from time to energy domain

$$G^R(E) = \int_{-\infty}^{\infty} d\tau e^{iE\tau/\hbar} G^R(\tau) = -i\int_0^{\infty} d\tau e^{i(E-H)\tau/\hbar}$$

The integral is oscillatory divergent as $\tau \to \infty$. To ensure convergence and satisfy the causality condition (*i.e.* poles lying in the lower half of the complex plane), introduce an infinitesimal positive real number η as an adiabatic switching factor $e^{-\eta\tau/\hbar}$

$$G^R(E) = -i\int_0^{\infty} d\tau e^{i(E-H+i\eta)\tau/\hbar} = -i\left[\frac{\hbar}{i(E-H+i\eta)} e^{i(E-H+i\eta)\tau/\hbar}\right]_0^{\infty} = \frac{\hbar}{E-H+i\eta} = \hbar(E-H+i\eta)^{-1}$$

Take the atomic unit: $\hbar = 1$

$$G^R(E) = (E-H+i\eta)^{-1}$$

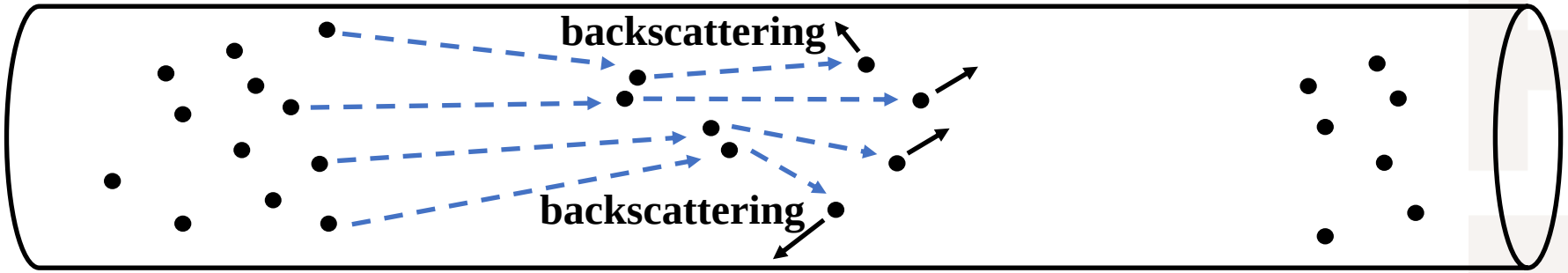
II. Spectral Dynamics and Quasiparticle Lifetime Analysis

$$G^R(E) - G^A(E) = -2\pi i A(k, E)$$

Spectral function: $A(k, E) = \frac{1}{\pi} \frac{|Im\Sigma|}{(E - H - Re\Sigma)^2 + (Im\Sigma)^2}$

In reality, electrons are always subject to scattering by phonons, impurities, or other electrons, resulting in a finite lifetime. In this case, the self-energy, $\Sigma = Re\Sigma + Im\Sigma$, is induced into the denominator of the Green's function

$$G^R(E) = (E - H + i\eta)^{-1}$$



$n > m$

ImΣ term, η, describe the lifetime of electrons due to the back scattering

Requirement for η to match the experimental scattering rate in physical systems with impurities

II. Spectral Dynamics and Quasiparticle Lifetime Analysis

$$G^R(E) = (E - H + i\eta)^{-1}$$

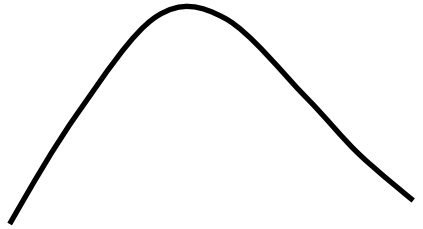
Effect of localized potentials in the Hamiltonian on the spectral function

Visualization of disorder smearing out sharp Van Hove singularities in the DOS

$\sigma_{\mu\nu}(\omega)$



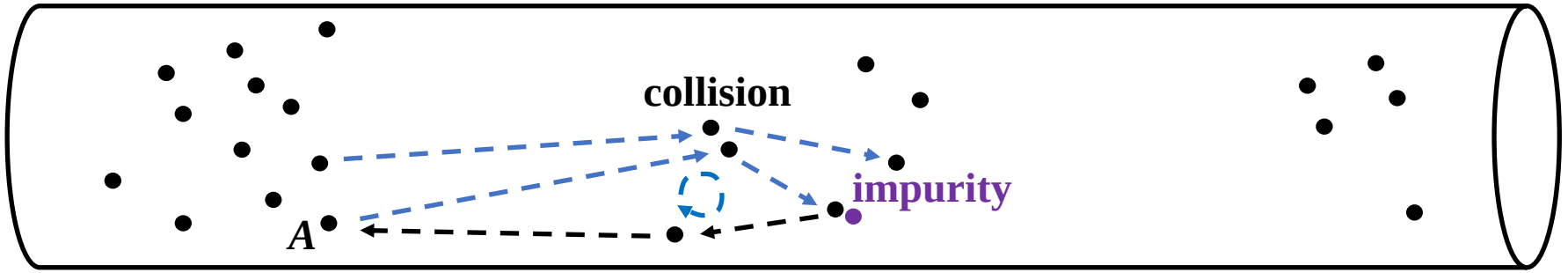
η – **Too small**



η – **Too large**

II. Spectral Dynamics and Quasiparticle Lifetime Analysis

Weak localization: Consider a specific scattering path that starts at point *A*, hits several impurities, and returns to point *A*. Because of time-reversal symmetry, the electron can traverse this same path in both the 'clockwise' and 'counter-clockwise' directions

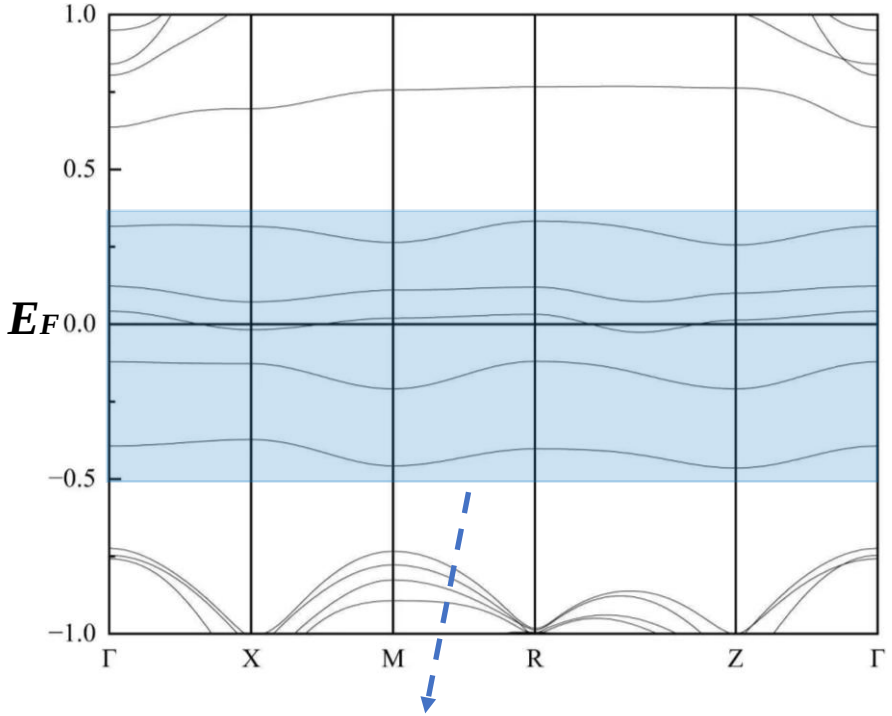


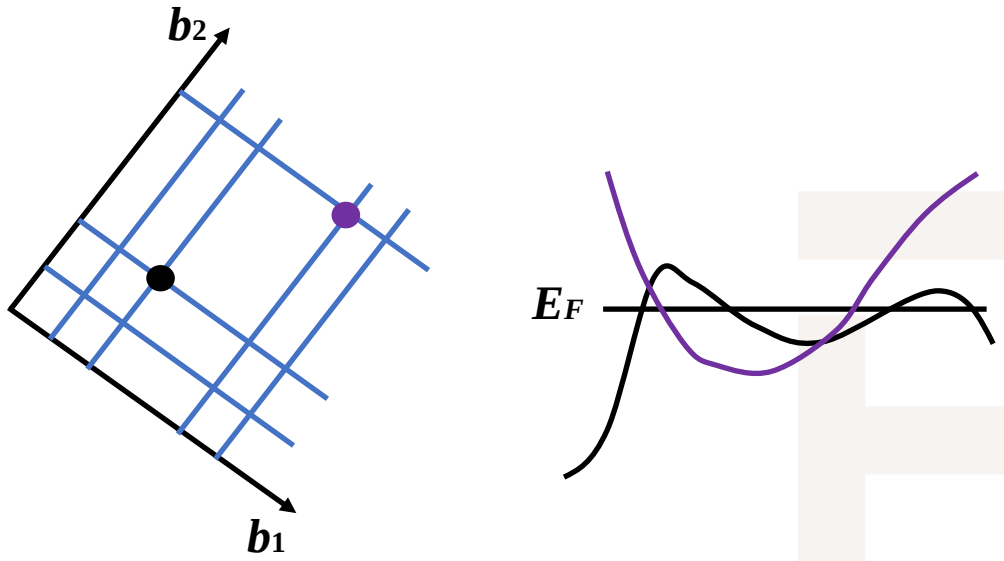
Constructive Interference: When these two waves (clockwise and counter-clockwise) return to the starting point, they are perfectly in phase with each other. This results in constructive interference, which enhances the probability of the electron being found back at its origin (point *A*)

The Kubo-Greenwood formula, however, explicitly includes the Green's functions (G^R and G^A). These functions contain information about the phase of the electron wave. When calculating the trace, the interference between all possible paths is mathematically accounted for, including these self-intersecting loops

II. Spectral Dynamics and Quasiparticle Lifetime Analysis

Kubo-Greenwood Formula $\sigma_{\mu\nu}(\omega) = \frac{\pi e^2 \hbar}{V} \int dE \left(-\frac{df}{dE}\right) Tr\{v_\mu [G_n^R(E) - G_n^A(E)] v_\nu [G_m^R(E) - G_m^A(E)]\}$





Reciprocal space

Infeasibility of evaluating σ at a single point E_F

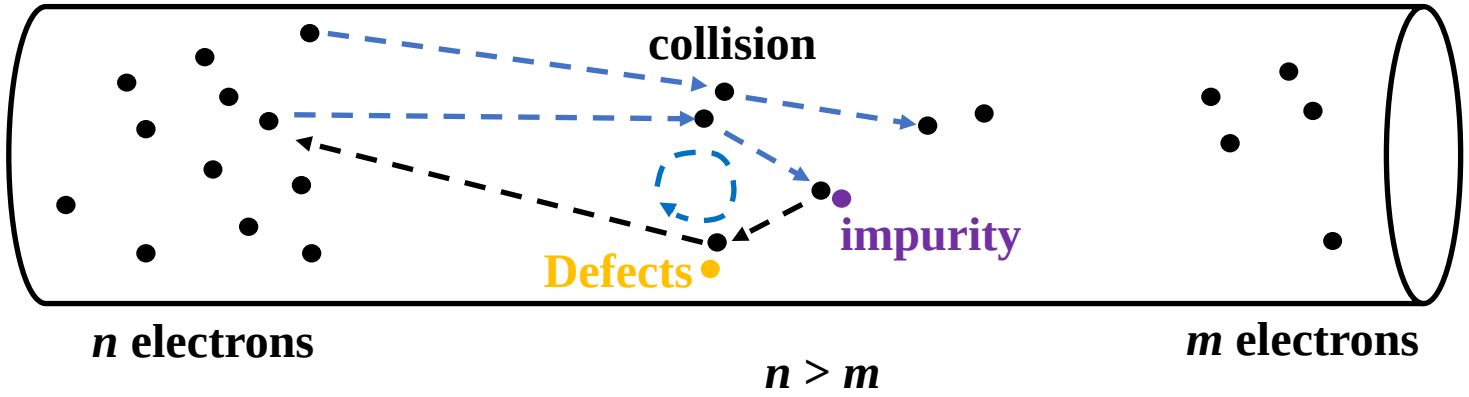
Necessity of an integration grid to capture the variation of the transport channel

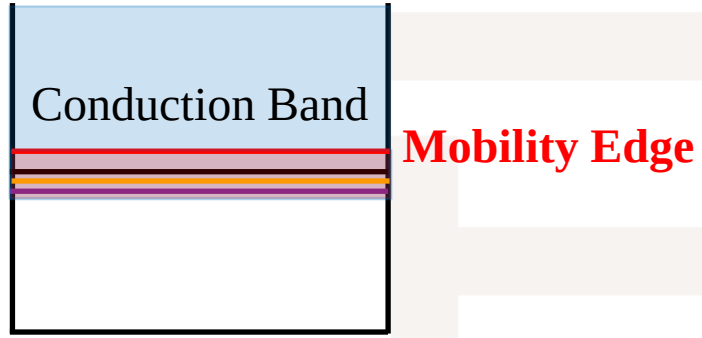
Decomposition of the Trace: Determining which energy states are 'conducting'

Identification of 'Hot Spots' in the Brillouin zone

II. Spectral Dynamics and Quasiparticle Lifetime Analysis

Anderson Localization: In a perfect crystal, the electron wavefunction extends throughout the entire material, allowing electrons to move freely — hence the metallic state. When disorder (impurities, defects) in the material becomes sufficiently strong, the electron wavefunction becomes localized within a finite spatial region, unable to propagate over long distances, and thus loses its ability to conduct, resulting in an insulating state



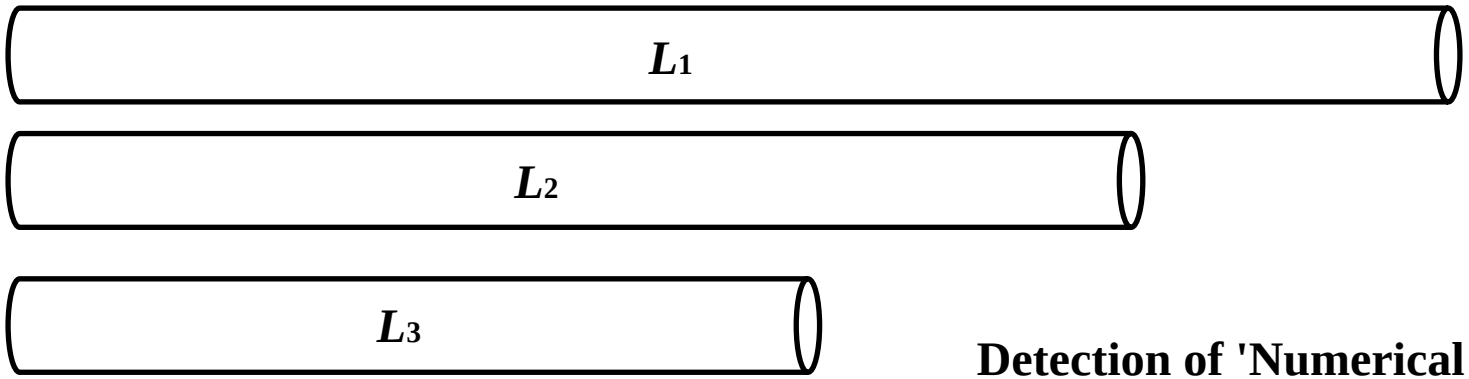


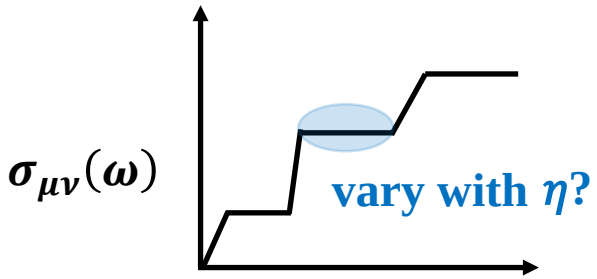
Mobility edge: In disordered materials, the fate of an electron depends on its energy. At a critical energy threshold known as the mobility edge, the nature of the electronic states changes fundamentally: above the mobility edge, the states are extended—the wavefunctions span the entire system, allowing for electrical conduction (metallic behavior); below the mobility edge, the states become localized—the wavefunctions are confined to a finite spatial region, unable to support long-range transport and thus contributing nothing to the DC conductivity (insulating behavior)

III. Numerical Diagnostics and Convergence in Computational Transport

Broadening and Lifetime Scaling: $\Delta E \approx \hbar/\tau$

Visualization of conductivity plateau changes with varying η



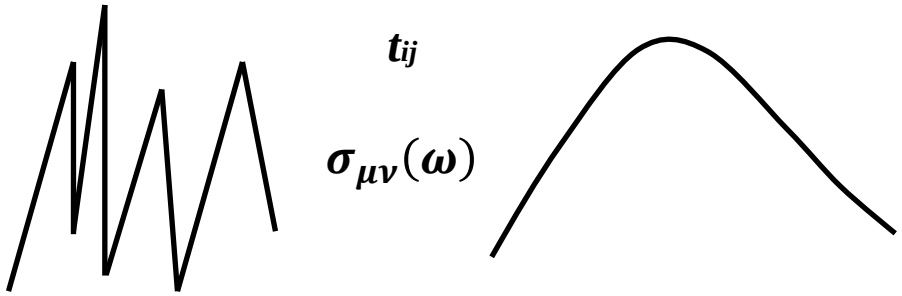


Detection of 'Numerical Noise' (indicative of insufficient η **or** L**)**

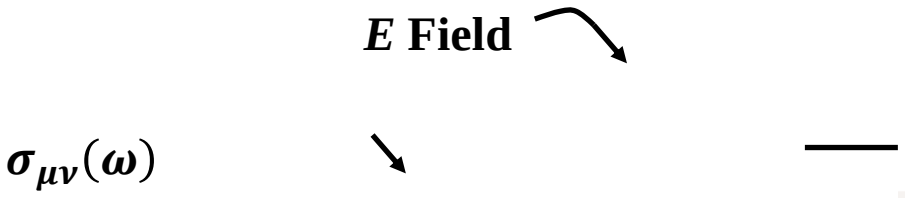
Differentiation between physical phenomena and numerical artifacts

III. Numerical Diagnostics and Convergence in Computational Transport

Examination of conductivity changes relative to hopping parameter (t_{ij}) perturbations
Importance of ensuring the 'weak field' assumption holds in linear response



Not converged converged

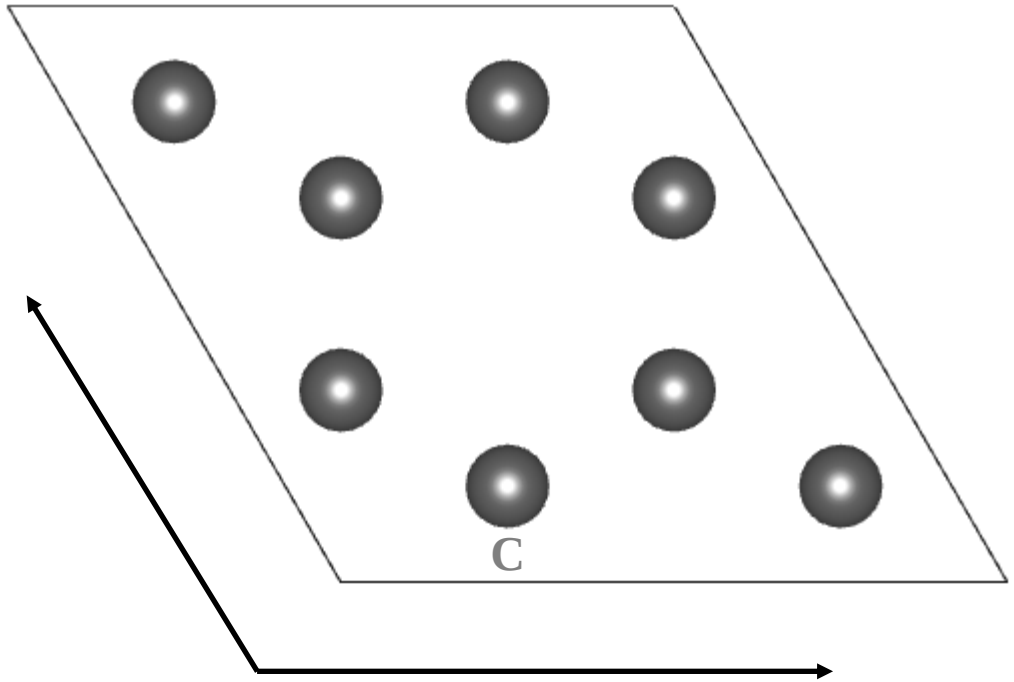


Non-Linear response Linear response

Topological insulators: Revelation of edge-state contributions through the KG formula.
Distinction between bulk and surface conductivity within the Trace formula

Observation	Physical Conclusion
Fermi energy lies in the bulk band gap, yet the total conductance is non-zero	Indicates the presence of conducting edge states
Fermi energy lies in the bulk band gap, and the total conductance increases with system size	Indicates that the edge states are metallic
Fermi energy lies in the bulk bands, but the surface contribution dominates over the bulk contribution	Indicates the system possesses topological surface states

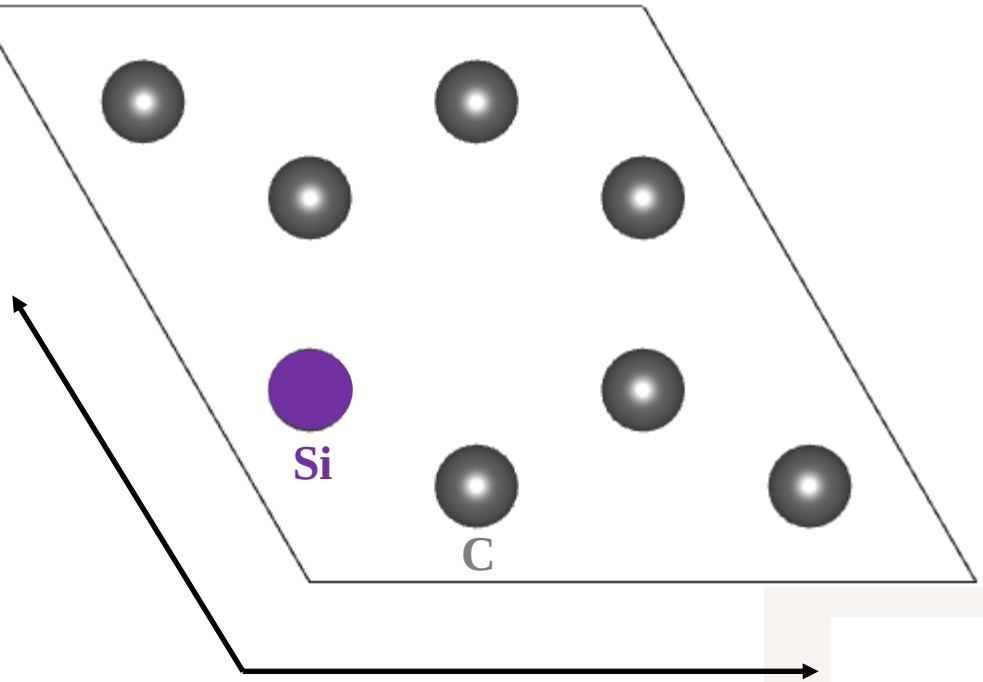
III. Numerical Diagnostics and Convergence in Computational Transport



Isotropy

$$v_x = v_y$$

$$\sigma_{xx} = \sigma_{yy}$$



Anisotropy

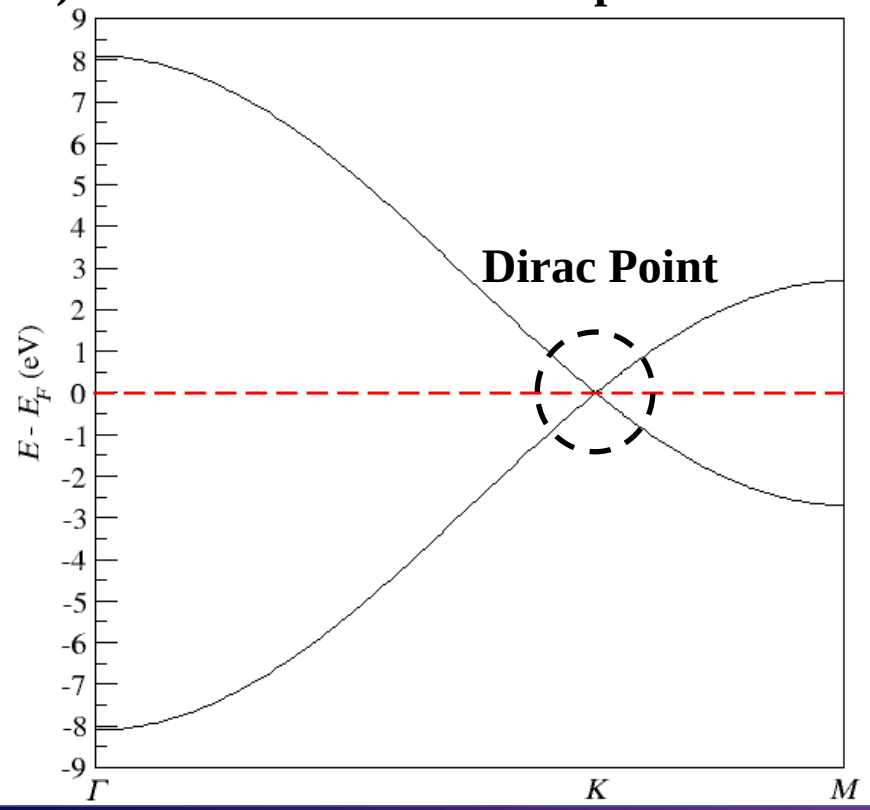
$$v_x \neq v_y$$

$$\sigma_{xx} \neq \sigma_{yy}$$

III. Numerical Diagnostics and Convergence in Computational Transport

In materials with a zero band gap, such as graphene, the Density of States (DOS) drops to zero at the Dirac point. Mathematically, this creates a potential 'singularity' in the Green's function calculations. When performing matrix inversion, *E*-*H* can become nearly zero, causing the numerical solver to struggle or produce values that explode (diverge)

The Solution: Introducing a finite η acts as a numerical regularizer. Physically, this represents the finite lifetime of the electron. It forces the system to acknowledge that even in an ideal crystal, electrons do not live forever; there is always some level of scattering (even if minimal). This stabilizes the computation and yields a physically realistic, finite conductivity value at the Dirac point



III. Numerical Diagnostics and Convergence in Computational Transport

♦ **Verification of Normalization: Ensure that the trace operation properly accounts for the system volume (Ω) and the number of states. Without this, the results are just relative numbers, not physical quantities (S/m)**

♦ **Convergence against System Size (L): Because conductivity is an intensive property, the result should not change significantly as the simulation cell grows larger. If the conductivity fluctuates wildly as L increases, the simulation has not yet reached the 'thermodynamic limit'.**

♦ **Physical Consistency of η: The broadening parameter must be 'tuned' to the physical reality of the sample. If a simulation assumes a ultra-pure crystal (η to 0) but attempts to model a dirty, polycrystalline sample, the results will be physically wrong. The parameter must match the expected scattering environment (*e.g.*, impurities, phonons, or defects).**

♦ **Final takeaway: The Kubo-Greenwood formula is not a 'magic button'. It is a delicate diagnostic tool. Take ownership of your computational results. Question your assumptions. Verify your convergence. That is what separates a code-executor from a true computational physicist.**



That's all for this session

Thank you for your attention and great participation